VW2-DirectAct Push-T Falsification

Oracle-Fix Final Report

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Verdict. After fixing rollout-side oracle conditioning, rerunning TeacherOracle, and sanity-checking the evaluator with a direct-act oracle policy, the Push-T continuous-subgoal branch still fails Gate A: **0.0% success on execute-1** and **0.0% success on execute-2**. StudentFrozen and StudentJoint were stopped.

Experiment ID. `pusht_subgoal_distill_round2_oraclefix`

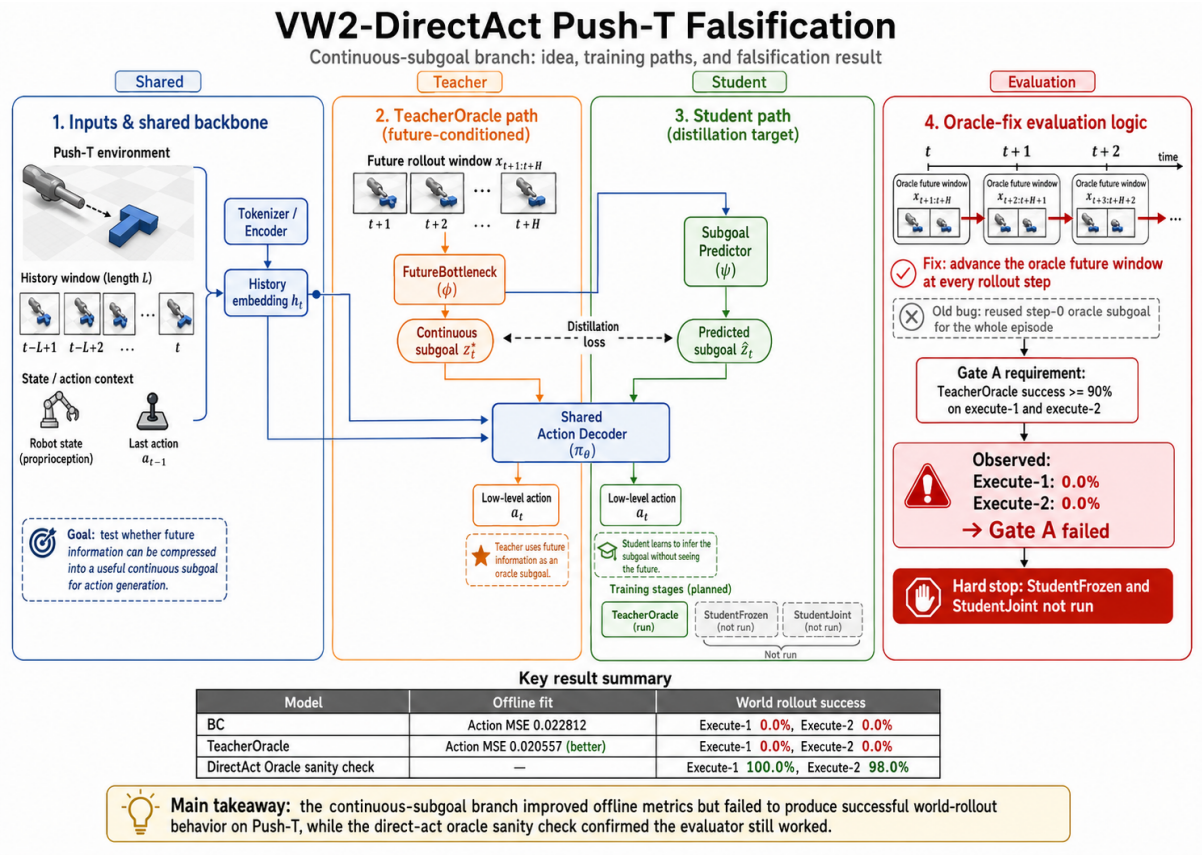

Figure 1: Overview illustration also used in the repository README. The source code, JSON/CSV artifacts, and this report are the source of truth for metrics and conclusions.

1. What Was Tested

Gate A asks whether future information can be compressed into a continuous subgoal that still drives successful Push-T world rollouts through the shared action decoder.

- **TeacherOracle**: encodes a future rollout window into a continuous subgoal.
- **Student**: predicts that subgoal from history; skipped because TeacherOracle failed Gate A.
- **Fix applied**: oracle future windows advance at every rollout step instead of reusing the step-0 oracle subgoal.

2. Evaluation Scope

Item	Value
Gate A rule	TeacherOracle success $\geq 90\%$ on execute-1 and execute-2
Rollouts	50 per execute setting, 100 world steps per rollout
Batching	`eval.rollout_batch_size=10`; memory-only change
Dataset starts	Deterministic starts from `pusht_expert_train.h5`
BC / direct-act starts	episodes 0–49, `start_step=0`
TeacherOracle starts	episodes 101–150, `start_step=3`
Generalization caveat	Offline splits are window-level, not episode-held-out splits

3. Results

Model	MSE	Exec-1	Reward-1	Exec-2	Reward-2
BC	0.02281195	0.0%	-23644.98	0.0%	-24199.06
TeacherOracle	0.02055714	0.0%	-21435.20	0.0%	-20406.16

TeacherOracle improves offline action MSE and mean reward over BC, but produces zero successes in 100 evaluated world rollouts. The branch fails on the behavior metric that matters for Gate A.

Evaluator sanity check. The fixed direct-act oracle evaluator still succeeds:

Model	Exec-1	Exec-2	Exec-4
DirectAct Oracle	100.0%	98.0%	0.0%

This separates evaluator correctness from the continuous-subgoal branch failure.

4. Evidence

- Artifact root: `artifacts/pusht_subgoal_distill_round2_oraclefix/`
- Main evidence: summary JSON, per-episode CSVs, rollout videos, TeacherOracle logs, and direct-act oracle sanity JSONs under the artifact root.
- Figures: shared overview `pusht_falsification_overview.png`; detailed diagrams in `docs/figures/`.
- Exclusions and guards: checkpoints, raw datasets, and local caches are excluded; world evaluation fails loudly when required rollout dependencies or starts are missing.